

Generating uncertainty
in process & outcome
ex: Coin flip & $\Omega = \{1, 2\}$
probability
 $P(W=H) = \frac{1}{\#|\Omega|} = \frac{1}{2}$
Equiprobable outcome models
like fair die $\Omega = \{1, 2, 3, 4, 5, 6\}$
but then frequentists came
& investigated events
w/ non-equiprobable outcomes
probability based on repeated trials
Trials = n
Favorable outcomes = S_n
as $n \rightarrow \infty$ frequency approaches $P(A)$
motivation for law of large numbers
asymptotic approach to "true" probability
Mathematicians (Gauss?)
said frequentists dealt only
in finite outcome spaces
for cont probability spaces
0 $\xrightarrow{\uparrow}$ 1
 $\xi \in \Omega$
 $P(A)$ & any
one outcome is zero
due to completeness
of real numbers
distinction b/w event $\xi \in \Omega$
w/ $P=0$ & event
outside domain, also $P=0$
 $P=0 \Rightarrow$ event is outside domain

Axioms:
Goal of prob theory
is to define function
 $P: \mathcal{P}([0,1]) \rightarrow [0,1]$ bounded
power set
A $\rightarrow P(A)$
event
 $P(\emptyset) = 0$ must hold
 $P([0,1]) = 1$
 $P(A^c) = 1 - P(A)$
 $\{A_i\}_{i=1}^\infty$ disjoint events
 $\Rightarrow P(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty P(A_i)$
Translation invariance
 $A \sim x+A$

Transcribed Notes: Foundations of Probability

Generating Uncertainty via Process and Outcome

- Example: Coin flip with sample space $\Omega = \{1, 2\}$
- Classical Probability:

$$P(\omega = H) = \frac{1}{\#\Omega} = \frac{1}{2}$$

- Equiprobable outcome models (e.g., fair six-sided die):

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Frequentist Perspective & Non-Equiprobable Outcomes

Frequentist formulation centers on empirical repeated trials:

- Total number of trials: n
- Number of favorable outcomes: S_n
- Asymptotic relative frequency:

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} \rightarrow P(S)$$

Motivation for the Law of Large Numbers:

Asymptotic convergence of sample frequencies to the underlying ("true") data-generating probability.

Theoretical Expansion: Epistemic & Measure-Theoretic Foundations

Classical Framework (Laplace)

[Finite Ω , Symmetric / Equiprobable]

▼ (Breakdown: Non-equiprobable / Continuous / Infinite)

Frequentist Formulation (von Mises)

[Empirical Limit of Relative Frequency]

▼ (Breakdown: Mathematical ill-definedness of "kollektiv")

Modern Axiomatic Framework (Kolmogorov, 1933)

[$(\Omega, \mathcal{F}, P)$ probability space]

Weak LLN (Chebyshev) Strong LLN (Kolmogorov)

$$S_n/n \xrightarrow{P} P(A) \quad S_n/n \xrightarrow{a.s.} P(A)$$

1. Classical Probability and the Principle of Indifference

The classical paradigm formalizes probability over a finite sample space under the **Principle of Insufficient Reason** (Laplace, 1814):

$$P(A) = \frac{|A|}{|\Omega|} = \sum_{\omega \in A} \frac{1}{|\Omega|}$$

Inherent Limitations:

- Circular Definition:** Defining probability via "equally likely" outcomes relies on an implicit, pre-existing concept of equal probability.
- Failure in Infinite Dimensions:** The uniform counting measure fails on infinite or non-compact sets, giving rise to geometric paradoxes (e.g., Bertrand's Paradox).

2. The Frequentist Program (von Mises)

Richard von Mises (1919) proposed grounding probability strictly in observable physical phenomena via infinite sequences of identical, independent experiments (termed a "kollektiv").

$$P(A)_{\text{kollektiv}} = \lim_{n \rightarrow \infty} \frac{S_n(A)}{n}, \quad S_n(A) = \sum_{i=1}^n \mathbf{1}_A(\omega_i)$$

Structural Defects of Frequentism as an Axiomatic System:

- Non-Constructive Limits:** An empirical limit cannot be tested or evaluated in finite time; actual experimental trials are finite $n < \infty$.
- Absence of Topological Structure:** In standard real analysis, a limit $\lim_{n \rightarrow \infty}$ requires a well-defined metric and a deterministic sequence (a_n) . Treating the sample path of a random experiment as a deterministic convergent sequence leads to mathematical contradictions unless constrained by algorithmic randomness (e.g., Martin-Löf randomness).

3. Kolmogorov's Measure-Theoretic Synthesis (1933)

Modern probability resolves this tension by treating probability not as a frequency limit, but as a normalized measure on a measurable space.

Definition (Probability Space):

A probability space is a triple $(\Omega, \mathcal{F}, P)$ where:

- Ω is an arbitrary sample space.
- $\mathcal{F} \subset 2^\Omega$ is a σ -algebra on Ω closed under complements and countable unions.
- $P: \mathcal{F} \rightarrow [0, 1]$ is a countably additive measure satisfying $P(\Omega) = 1$.

4. Rigorous Recovery of Frequentist Intuition: Laws of Large Numbers

Under Kolmogorov's system, empirical convergence is not an axiom or definition; it is a **derived analytical theorem** resulting from the interaction of measure theory and product spaces $(\Omega^n, \mathcal{F}^n, P^n)$.

Let $X_i = \mathbf{1}_A$ be independent and identically distributed (i.i.d.) indicator random variables with $\mathbb{E}[X_i] = P(A)$ and $\text{Var}(X_i) = P(A)(1 - P(A)) < \infty$. Let $S_n = \sum_{i=1}^n X_i$.

A. Weak Law of Large Numbers (WLLN)

Convergence in Probability: For every $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{S_n}{n} - P(A)\right| \geq \epsilon\right) = 0$$

Proof via Chebyshev's Inequality:

$$\text{Var}\left(\frac{S_n}{n}\right) = \frac{\text{Var}(X_1)}{n} = \frac{P(A)(1 - P(A))}{n} \xrightarrow{n \rightarrow \infty} 0$$

$$P\left(\left|\frac{S_n}{n} - P(A)\right| \geq \epsilon\right) \leq \frac{\text{Var}(S_n/n)}{\epsilon^2} = \frac{P(A)(1 - P(A))}{n\epsilon^2} \xrightarrow{n \rightarrow \infty} 0$$

B. Strong Law of Large Numbers (SLLN)

Almost Sure Convergence:

$$P\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} \frac{S_n(\omega)}{n} = P(A)\right\} \right) = 1$$

The SLLN establishes that the set of sample paths whose empirical relative frequency fails to converge to the measure-theoretic probability $P(A)$ has P^n -measure zero.

From: <https://papers.ssrn.com/sol3/cfdoc029784840839/>

Probability Axioms and Translation Invariance

Goal of Probability Theory:

To define a function mapping from the powerset of $[0,1]$ to a bounded output between 0 and 1:

$$P: \mathcal{P}([0,1]) \rightarrow [0,1]$$

where $A \mapsto P(A)$ for an event A .

Sidebar Definition: An event A is a set of primary outcomes from the sample space Ω .

(Accompanied by a Venn diagram showing disjoint events A_i and A_j within Ω .)

Required Axioms and Properties:

- Null Event:** $P(\emptyset) = 0$ must hold.
- Normalization:** $P([0,1]) = 1$.
- Complement:** $P(A^c) = 1 - P(A)$.
- Countable Additivity:** For a sequence of disjoint events $\{A_i\}_{i=1}^\infty$ where $A_i \cap A_j = \emptyset$:

$$P\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty P(A_i)$$

Translation Invariance:

Shifting an event along the domain (e.g., $x \mapsto x + A$) does not change its probability:

$$P(x + A) = P(A)$$

Analytical Expansion: The Impossibility of the Powerset Domain

Your notes outline the intuitive goal of assigning a probability to every possible subset of the continuous domain $[0,1]$ (the powerset $\mathcal{P}([0,1])$). However, a foundational result in measure theory proves that **this specific mathematical goal is strictly impossible**.

You cannot simultaneously have a function that satisfies:

- Defined on the entire powerset $\mathcal{P}([0,1])$
- Countably additive
- Translation invariant
- Normalized ($P([0,1]) = 1$)

The Vitali Paradox (1905)

If you assume such a measure P exists on the entire powerset of $[0,1]$, you can construct a "non-measurable" set that breaks mathematics.

1. Construction (using the Axiom of Choice):

Define an equivalence relation on $[0,1]$ where two numbers are equivalent if their difference is a rational number ($x \sim y \Leftrightarrow x - y \in \mathbb{Q}$). Partition $[0,1]$ into these equivalence classes. Now, form a new set V (the Vitali set) by picking exactly **one** representative number from each equivalence class.

2. The Problem of Disjoint Sets:

Transcribed Notes: Continuous Probability Spaces and Measure Zero

Critique of Early Frameworks:

Mathematicians (Blue query: "Gauss?") noted that frequentists dealt primarily with finitely discrete outcome spaces.

Continuous Probability Spaces:

Consider the unit interval $[0,1]$:

- For any single singleton outcome $\{a\} \subset [0,1]$:

$$P(\{a\}) = 0$$

due to the uncountable continuum / completeness of real numbers.

Distinction Between Null Events and Impossible Events:

- An event $\{a\} \subset [0,1]$ inside the domain with $P(\{a\}) = 0$.
- An event entirely outside the sample space domain Ω , which also has $P = 0$.
- Fundamental implication:

$$P(E) = 0_{\text{centered}} \Rightarrow E \text{ is impossible / outside domain}$$

Response to Sidebar Question & Analytical Expansion

1. Sidebar Question: "Gauss?"

While Carl Friedrich Gauss contributed extensively to the normal error distribution and least squares (early 1800s), the rigorous critique regarding infinite/continuous outcome spaces and the inability of classical/early frequentist formulations to handle continuous geometries traces historically to:

- Joseph Bertrand (1889):** Formulated Bertrand's Paradox, demonstrating that assigning probabilities uniformly over continuous geometries is ill-defined without a specifying measure.
- David Hilbert (1900):** Hilbert's Sixth Problem explicitly tasked mathematicians with axiomatizing probability theory using analytical and measure-theoretic mechanics, which culminated in **Andrey Kolmogorov (1933)**.

2. Theoretical Expansion: Why Kolmogorov Resolves This

This note directly captures the primary motivation behind Kolmogorov's measure-theoretic framework $(\Omega, \mathcal{F}, P)$: the transition from discrete counting measures to continuous probability spaces.

Event Possibility vs. Measure

Impossible Event: $E = \emptyset$	$\longrightarrow P(\emptyset) = 0$ (Strict Axiom)
Null (Possible): $\{a\} \subset [0,1]$	$\longrightarrow P(\{a\}) = 0$ (Measure-Theoretic Property)
Almost Sure $\{a, \omega\} \subset A \subset [0,1]$ (a)	$\longrightarrow P(A) = 1$ (Yet $A \neq \Omega$)

A. The Continuum & Countable Additivity

If $\Omega = [0,1]$ endowed with the uniform Lebesgue measure λ , the probability of an interval is $P([c,d]) = d - c$.

- For any singleton $\{a\} = [a, a]$:

$$P(\{a\}) \leq P\left(\left[a - \frac{1}{n}, a + \frac{1}{n}\right]\right) = \frac{2}{n} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow P(\{a\}) = 0$$

- If one attempted to sum singletons to get the probability of $[0,1]$, the classical sum $\sum_{x \in [0,1]} P(\{x\}) = \sum_{x \in [0,1]} 0 = 0 \neq 1$ breaks down.
- Kolmogorov resolved this by restricting additivity to **countable** collections (σ -additivity). Because $[0,1]$ is uncountably infinite, σ -additivity does not apply to arbitrary uncountably infinite unions of disjoint singletons.

B. Null Sets vs. The Empty Set

- Impossible Event:** $E = \emptyset$, for which Kolmogorov's axioms strictly enforce $P(\emptyset) = 0$.
- Null Event (Event of Measure Zero):** A non-empty event $A \in \mathcal{F}$ ($A \neq \emptyset$) such that $P(A) = 0$. In a continuous distribution, every singleton $\{a\}$ is possible (it can occur as the realized sample path), yet its measure is zero.
- Almost Sure Event ($P(A) = 1$):** An event E occurs almost surely if $P(E) = 1$. This does not mean $E = \Omega$; non-occurring outcomes can exist in $\Omega \setminus E$, provided the set of exceptions has measure zero.

From: <https://papers.ssrn.com/sol3/cfdoc029784840839/>

Why?
Where is contradiction?

Vitali proposition is that
it is impossible to have
IP measure that satisfies
all of the above

Solution may be restricted domain rather than powerset over $[0,1]$, use sigma algebra

Def of sym algebra S
(or sym field)

Ω is set of possible atoms
Subsets of Ω are events A

σ algebra is family of subsets

$$F \subseteq \overset{\text{powerset}}{\mathcal{P}(\Omega)} \text{ for } F \neq \emptyset$$

$$1. A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$$

2. $\{A_i\}_{i=1}^{\infty}$ $A_i \in \mathcal{F}$

$$\Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$$

Can we
measure infinitely
infinitesimal events?

Condition 1 & 2, $\Omega \in \mathcal{F}$

für $A \in \mathcal{F}$ und $A^c \in \mathcal{F}$,

by 2 : $A \cup A^c \cup \emptyset = \Omega \in \mathcal{F}$

does A_i^c include all A_j $A_j \neq i$?

So the $\emptyset$ is Ω^c .

$$\emptyset \in \mathcal{F}$$

$$\text{Take } \bigcap_i A_i = (\bigcup_i A_i^c)^c$$

Following De Morgan's
 $(A \cap B)^c = A^c \cup B^c$
 $\downarrow$
 $\cup \cap$

shy I closed under
comfortable unions

What is σ algebra for?
 notion of information,
 what is knowable

ex: $\Omega = \{1, i\}$

$$F = \{\emptyset, \Omega, \{H\}, \{T\}\}$$

once event observed,
complement cannot happen
so event gives
information on complement

2. Sub- σ -Algebras and Measurability

A sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$ represents **coarser / partial information**:

- A random variable $X: \Omega \rightarrow \mathbb{R}$ is **measurable with respect to** $\mathcal{G}$ (written $X \in \mathcal{G}$) if the value of X can be determined solely from the events decidable in $\mathcal{G}$:

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad X^{-1}(B) \in \mathcal{G} \iff \{\omega \in \Omega: X(\omega) \in B\} \in \mathcal{G}$$

- In stochastic processes, an increasing sequence of σ -algebras $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}$ is called a **filtration**, mathematically encoding the flow of information accumulating over time.

From <https://www.google.com/maps/@40.7269748,-85.8283583>